SO(k)-Invariant Gauge Rigidity for Declared Euclidean Stalks

Local ℓ^2 Selection and Outer ℓ^1 Aggregation (GS-002)

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Abstract

We study vector-valued defect diagnostics on finite directed graphs with declared Euclidean stalks. The classical sphere-transitivity argument shows that a faithful absolutely homogeneous $\mathrm{SO}(k)$-invariant local gauge is a positive multiple of $\|\cdot\|_2$. Substitution into a separately declared strong edgewise sum gives $J_{2,1,c}(\omega) = \sum_e c_e \|\omega_e\|_2$; symmetry alone does not determine the outer aggregation. Edge exchange symmetry makes weights constant on declared orbits, while unit normalization fixes them to one.

For identity transports, finite-dimensional convex duality identifies the quotient norm modulo vertex repairs with a maximum over divergence-free vector witnesses satisfying $\|\varphi_e\|_2 \le c_e$. We give an attained single-cycle formula $(\min_e c_e)\|S(\omega)\|_2$, a normalized triangle separation of 2 versus $\sqrt{2}$, and a closure bound under $d_1 d_0 = 0$. These are applications of standard norm, cycle-space, and convex-duality results with explicit observer hypotheses. The companion Lean development mechanizes finite and abstract kernels, including planar rotation transitivity and weak duality; general $\mathrm{SO}(k)$ transitivity and strong duality remain prose or imported steps. Python examples evaluate analytical formulas and numerical certificates in floating-point arithmetic. Finite here describes the carrier and stalk dimension; the Euclidean scalar and symmetry structures are declared continuum inputs.

> **What this does not claim.** This paper does not claim global Euclidean physics, global ℓ^1 geometry, unweighted $\ell^1(E;\ell^2)$ without edge normalization, or that every vector-valued observer must use $\mathrm{SO}(k)$. It does not claim that the triangle example shows the general theory by itself. It claims only that, under declared $\mathrm{SO}(k)$-invariant local gauge symmetry and exact edge-additive replica extensivity, the licensed finite diagnostic is weighted $\ell^1(E;\ell^2)$, with unweighted form requiring normalization or edge-uniformity.
>
> **Why a researcher should care.** Many finite obstruction systems have vector-valued local defects: displacement fields, force/strain residuals, multi-channel sensor mismatches, projected dynamics errors, and vector-valued presheaf gluing failures. Treating such a defect as independent scalar coordinates gives a different diagnostic from treating it as one Euclidean vector. This paper identifies the precise observer assumption that justifies the Euclidean treatment: continuous $\mathrm{SO}(k)$ rotational invariance of the local gauge. It then separates that local symmetry choice from the outer edge aggregation assumption, showing that $\ell^1(E;\ell^2)$ is a conditional observer geometry, not a global metric.

1. Introduction and Structural Placement

The classification of diagnostic geometries over finite graphs forms the basis of categorical obstruction analysis. Previous classification theorems show that when an observer's operational domain demands edge-decomposable aggregation and maximal bounded-linear compatibility on scalar stalks, the resulting diagnostic is a weighted ℓ^1 coboundary norm, becoming uniform only under edge-uniformity assumptions.

However, moving from scalar signals to Euclidean geometries requires stating the structural constraints governing vector variables. Evaluating a 2D vector defect $\vec{\delta} = (\delta_x, \delta_y)$ via $|\delta_x| + |\delta_y|$ treats

the two coordinates as independently observed scalar channels.

Conversely, treating $\vec{\delta}$ as a single vector under continuous rotation selects the local stalk gauge $\|\vec{\delta}\|_2$, up to scale.

The aggregate measuring these local bounds independently across edges is then the mixed $\ell^1(E;\ell^2)$ measure. This work formalizes the geometric separation between coordinate-separable scalar error components and vector-valued Euclidean defects.

Falsifiability. A proposed use of this paper fails if it cannot specify the finite carrier, vector stalk dimension $k \geq 2$, local $\mathrm{SO}(k)$-invariant gauge assumption, edge-decomposable and disjoint-additive aggregation assumptions, positive edge weights c_e, and the normalization or edge-orbit condition used to pass from weighted to unweighted $\ell^1(E;\ell^2)$. The duality claim fails if a finite counterexample satisfies the primal quotient problem but violates the stated divergence-free Euclidean-ball dual formula.

1.1 Related Work and Positioning

This paper is closest to five bodies of work:

1. **Invariant norm and convex-body classification.** The $\mathrm{SO}(k)$ result is a special case of norm classification under group actions: transitivity on the Euclidean unit sphere forces an invariant norm to be Euclidean up to scale (a classical fact of convex geometry / normed-space theory — a finite-dimensional norm whose linear isometry group acts transitively on the unit sphere is induced by an inner product). This paper differs by applying that local classification to finite obstruction cochains.
2. **Banach-space direct sums and mixed norms.** The resulting diagnostic is a mixed norm: ℓ^1 across edges and ℓ^2 inside each vector stalk. Mixed $\ell^p(\ell^q)$ spaces are standard in functional analysis (e.g., Benedek–Panzone-type mixed-norm spaces). This paper differs by deriving the inner/outer split from separate observer assumptions rather than choosing the mixed norm as a modeling convention.
3. **Finite cochain and quotient-norm obstruction theory.** The primal obstruction is a quotient norm modulo exact repairs. This paper differs by specializing the cochain geometry to $\mathrm{SO}(k)$-invariant vector stalks and deriving the corresponding dual witnesses.
4. **Convex duality / Fenchel duality and network flow.** The mixed duality theorem is a finite-dimensional convex-duality result: the dual of $c_e\|\cdot\|_2$ is the Euclidean ball $\|\varphi_e\|_2 \leq c_e$ by Fenchel–Rockafellar convex duality. The divergence-free constraint $d_0^*\varphi = 0$ makes the dual a vector-valued network-flow / min-cost-circulation problem. This paper differs by interpreting those dual variables as divergence-free vector circulations.
5. **Vector-valued sensor, strain, and displacement diagnostics.** Many applied systems are *modelled* with Euclidean vector defects rather than coordinate-separable scalar errors. This paper differs by giving the finite observer assumptions under which that Euclidean local geometry is licensed. That such systems *are* Euclidean is a modelling declaration, not a finding of this paper; the applied vocabulary in §1 and §5 names interpretation candidates, not realized physics (register row **I8**).

These are positioning anchors. The cited external results are used only for background context unless explicitly invoked in a theorem.

1.2 Evidence and Computational Scope

The companion claim ledger distinguishes local proofs, imported results, and partial mechanizations. The Lean radial-collapse theorem uses a declared ambient norm and assumes sphere constancy; the explicit planar circle-transitivity result is a separate algebraic theorem. The finite cycle repair kernel uses an anchored presentation, and the full graph-incidence identification is supplied by the prose argument in §5. Strong duality and attainment in §4 use standard finite-dimensional convex analysis, beyond the Lean weak-duality kernel.

The executable graph engine implements nonempty finite carriers with identity transports, common vector dimension, and scalar signed face incidence. Its analytical cycle formula is exact as mathematics, but its numerical evaluation and residuals use floating-point arithmetic. A numerical `is_optimal` flag means that the reported gap, divergence, and ball violation pass the declared tolerance; it is not an exact-arithmetic theorem certificate. Rotation and sphere audits test finitely many samples and cannot establish unbounded invariance. Fresh build and notebook replay evidence is recorded separately from retained historical outputs in `docs/publication_review.md`.

2. General M-Compatible Gauge Classification

Evaluating obstruction geometries over finite carriers requires mapping local measurement constraints onto an admissible transformation monoid.

2.1 Definition

Let V be a finite-dimensional normed vector space and let

$$M \subset \mathrm{End}(V)$$

be an admissible transformation monoid representing local observer symmetries.

A faithful gauge $j : V \to \mathbb{R}_{\geq 0}$ is **M-compatible** if:

1. $j(x) = 0 \iff x = 0$
2. $j(\lambda x) = |\lambda| j(x)$
3. $j(x + y) \leq j(x) + j(y)$
4. $j(Tx) \leq \|T\|_{\|\cdot\|_V \to \|\cdot\|_V} j(x)$ for all $T \in M$, where $\|\cdot\|_V$ is the declared ambient norm.

Equivalently, j is a norm whose unit ball is compatible with the action of M.

> **[IMPORTED EVIDENCE FROM THE M-COMPATIBLE GAUGE PAPER; NOT LOCALLY PROVED] Theorem 2.1 — General M-Compatible Gauge Classification**
>
> Let V be finite-dimensional and $M \subset \mathrm{End}(V)$ an admissible transformation monoid. There is a one-to-one correspondence between M-compatible faithful gauges j and closed, convex, balanced, absorbing, **bounded** unit bodies $B \subset V$ satisfying
>
> $$T(B) \subset \|T\| B \quad \text{for all } T \in M.$$
>
> The correspondence is given by the Minkowski functional $j_B(x) = \inf\{\lambda > 0 : x \in \lambda B\}$.
>
> Consequently, classifying M-compatible obstruction theories is equivalent to classifying M-contractive symmetric convex bodies. Increasing M shrinks the admissible body class. The maximal case $M = \mathrm{End}_b(V)$ collapses the body class to scalar multiples of the native Banach unit ball, recovering a unique local norm (up to scale); ℓ^1 arises only after imposing replica-extensive aggregation across edges.

This statement is reproduced solely as an imported dependency. This paper supplies neither a local proof nor a Lean mechanization of Theorem 2.1; any use below is conditional on the cited companion source and inherits its claim status.

Boundedness is essential and matches the companion source's finite-dimensional statement. For example, $B = V$ is closed, convex, balanced, and absorbing, but its Minkowski functional is identically zero and is not faithful. In finite dimension a closed balanced convex absorbing body is a neighborhood of zero; boundedness excludes nonzero directions with zero gauge.

2.2 Corollaries

- **[IMPORTED FROM THE M-COMPATIBLE GAUGE PAPER / PROVEN] Corollary 1 (Order Reversal):** If $M_1 \subset M_2$, then $\mathrm{Gauge}(V, M_2) \subset \mathrm{Gauge}(V, M_1)$. More symmetry weakly reduces the admissible gauge class; strict reduction requires a separating example.
- **[IMPORTED FROM THE M-COMPATIBLE GAUGE PAPER / PROVEN, CONDITIONAL + Hahn–Banach / native norm] Corollary 2 (Maximal Symmetry):** If $M = \mathrm{End}_b(V)$, then $\mathrm{Gauge}(V, M) = \{c\|\cdot\| : c > 0\}$. This recovers the maximal-local-symmetry endpoint used by the companion gauge-classification paper.
- **[PROVEN, CONDITIONAL + isometric group action] Corollary 3 (Group Symmetry & Native Norm Rigidity):** If $M = G$ is a group of linear isometries on a normed space $(V, \|\cdot\|_V)$, then $j(gx) = j(x)$ for all $g \in G$, and the M-compatible gauges are exactly the G-invariant norms. Equivalently, the unit body B_j is a G-invariant symmetric convex body. If G acts transitively on the native unit sphere $\{x \in V : \|x\|_V = 1\}$, then positive homogeneity and faithfulness determine the local gauge up to scale as $j(x) = c\|x\|_V$ for some $c > 0$. In particular, when $V = \mathbb{R}^k$ is equipped with the standard Euclidean inner product and $G = \mathrm{SO}(k)$, this specializes to $j(x) = c\|x\|_2$.
- **[IMPORTED FROM THE M-COMPATIBLE GAUGE PAPER / STANDARD / PROVEN, CONDITIONAL + compact group + Haar measure] Corollary 4 (Compact Group Construction):** If G is compact, acts continuously and linearly, and q is any norm, then $j_G(x) := \int_G q(gx)dg$ is a G-invariant norm. Thus, compact symmetry classes generally admit vast continuous families of gauges unless the group acts transitively enough to collapse them.
- **[STANDARD / SUPPORT LEMMA] Support Lemma S8 (Orbit Convex Hull Description):** For a compact group G acting continuously and linearly on finite-dimensional V and a bounded seed set S spanning V, the closed balanced convex hull $B = \overline{\mathrm{aconv}}(\bigcup_{g\in G} gS)$ is bounded, absorbing, and G-invariant. Its Minkowski functional is a faithful G-invariant norm. If S is already balanced, the ordinary closed convex hull suffices.

Warning: There is generally **no finite list** of M-compatible gauges. The classification maps $M \to$ family of invariant/contractive convex bodies. Only restrictive maximal monoids collapse the hierarchy to a single local gauge class.

3. Formal Framework: Directed Graph Complex and $\mathrm{SO}(k)$ Geometry

Let $G = (V, E, s, t)$ be a finite directed graph with vertex set V, an indexed edge set E, and source and target maps $s, t : E \to V$. Indexed edges allow parallel comparisons with the same endpoints. We write $e = (u \to v)$ when $s(e) = u$ and $t(e) = v$.

3.1 [DEFINITION] The Vector-Valued Cochain Spaces

Each vertex $v \in V$ carries a vector assignment $a(v) \in \mathbb{R}^k$. For each directed edge $e = (u \to v) \in E$, declare transport $r_{v,u} : \mathbb{R}^k_u \to \mathbb{R}^k_v$ in $\mathrm{GL}(k, \mathbb{R})$. The target-stalk comparison defect is

$$\delta_e(a) = a(v) - r_{v,u} a(u)$$

where $\delta_e \in \mathbb{R}^k$.

In the standard flat reference setting $r_{v,u} = I$, the coboundary operator $d_0 : C^0(G; \mathbb{R}^k) \to C^1(G; \mathbb{R}^k)$ acts on vertex fields $\alpha \in C^0(G; \mathbb{R}^k)$ by:

$$(d_0 \alpha)_e = \alpha(v) - \alpha(u).$$

The graph $G = (V, E)$, the edge transports, and the field assignments are declared combinatorial data. They provide an explicit combinatorial index for exhibited edge comparisons rather than presupposing a continuous ambient embedding, an antecedent physical space, or an unindexed observer. Common over-interpretations are explicitly scoped out:

1. G is a declared combinatorial complex indexing comparisons, not an unindexed spatial continuum.
2. The assignment a parameterizes local vertex values; δ_e is an exhibited cochain defect.
3. $\mathrm{SO}(k)$ is a declared local gauge group acting on stalk fibers, not an external coordinate rotation of physical space.

3.2 Local Gauge Theorem

[PROVEN, CONDITIONAL + $k \geq 2$, faithful absolute homogeneous convex gauge] Theorem 3.1 — $\mathrm{SO}(k)$ **Invariance Forces the Euclidean Local Gauge.** Let $k \geq 2$ and let $j : \mathbb{R}^k \to \mathbb{R}_{\geq 0}$ be a faithful absolute homogeneous convex gauge satisfying

$$j(Rx) = j(x) \qquad \forall R \in \mathrm{SO}(k),\ x \in \mathbb{R}^k.$$

Then there is a constant $c > 0$ such that

$$j(x) = c\|x\|_2 \qquad \forall x \in \mathbb{R}^k.$$

The $k = 1$ case is the scalar one-dimensional case: every faithful homogeneous norm is proportional to absolute value. The SO(k) transitivity proof is only needed for $k \geq 2$.

Proof. The group $\mathrm{SO}(k)$ acts transitively on every Euclidean sphere

$$S^{k-1}_r = \{x \in \mathbb{R}^k : \|x\|_2 = r\}$$

for $k \geq 2$. Therefore j is constant on each sphere. Define

$$\phi(r) := j(x) \quad \text{for any } x \text{ with } \|x\|_2 = r.$$

This is well-defined by $\mathrm{SO}(k)$ invariance. By positive homogeneity of the gauge,

$$\phi(r) = j(ru) = r j(u)$$

for any Euclidean unit vector u. Since j is constant on the unit sphere, $j(u) = c$ for all $\|u\|_2 = 1$. Faithfulness gives $c > 0$. Hence

$$j(x) = c\|x\|_2$$

for all $x \neq 0$, and the formula also holds at $x = 0$. □

This theorem is local. It selects the Euclidean stalk gauge only after the observer has declared full continuous rotational symmetry. Signed-permutation symmetry alone would not select ℓ^2; it admits many unconditional norms.

The radial argument itself uses only invariance, nonnegativity, faithfulness, and positive homogeneity; convexity and subadditivity are not needed for this step. They remain part of the norm hypotheses used for cycle minimization and convex duality below.

IMPORT — the inner product is admitted together with $\mathrm{SO}(k)$; it is not constructed at this degree.

$\mathrm{SO}(k) = \{R \in GL(k,\mathbb{R}) : R^\top R = I,\ \det R = 1\}$ is *definitionally* the isometry group of the standard inner product on $\mathbb{R}^k$: the condition $R^\top R = I$ cannot be stated without that inner product, and the proof above quantifies over the Euclidean spheres $\|x\|_2 = r$ — the selected norm appears in the proof of its own selection. **Declaring $\mathrm{SO}(k)$ is therefore declaring the inner product.** Theorem 3.1 is a *rigidity* theorem relative to an already-admitted structure: a gauge declared invariant under the isometry group of an admitted inner product cannot differ from that inner product's norm, up to scale. It is **not** a construction of the ℓ^2 fibre at this paper's declared degree, and it does not make that fibre cheaper than the structure it arrives with. See the *Import / Oracle boundary* register, rows **I1**–**I3**.

The theorem's mathematical force is real and is **not** withdrawn: among all faithful absolutely homogeneous convex gauges on $\mathbb{R}^k$, the $\mathrm{SO}(k)$-invariant ones are exactly the positive multiples of $\|\cdot\|_2$ — a genuine, non-vacuous restriction of the gauge class. What is priced here is the *direction of supply*, not the validity of the implication.

The paper's own mechanization exhibits this boundary rather than hiding it. In `lean4/gs002_l003_02_s_ok_local_gauge.lean`, while an explicit constructive SO(2) circle-transitivity theorem (`so2_transitive_on_circle`) is formalized for $\mathbb{R}^2$, the general k-dimensional theorem `so_k_local_gauge_simplified` is stated over an arbitrary `NormedAddCommGroup V`: it *assumes* unit-sphere constancy rather than constructing a general Lie group $\mathrm{SO}(k)$ action on $\mathbb{R}^k$, and concludes $j = c\|\cdot\|$ for **the declared ambient norm** — instantiated at $(\mathbb{R}^2, \ell^1)$ it returns $j = c\|\cdot\|_1$. The Euclidean content for general k is carried entirely by the choice of ambient norm, i.e. by the import. This note records what it costs.

3.3 Edge Aggregation and Normalization

[CONDITIONAL / DEFINITIONAL AGGREGATION; local rigidity supplied by Theorem 3.1] Theorem 3.2 — Weighted Mixed Norm Under Declared Strong Edge Aggregation. Let P be a finite directed graph with edge defects $\omega_e \in \mathbb{R}^k$, $k \geq 2$. Suppose:

1. each local gauge j_e is faithful, homogeneous, convex, and $\mathrm{SO}(k)$-invariant;
2. the global diagnostic is declared strongly edge-decomposable, meaning $J(\omega) = \sum_{e\in E} j_e(\omega_e)$;
3. this declared edgewise sum is additive over disjoint replicas.

Then there are positive edge weights $c_e > 0$ such that

$$J_{2,1,c}(\omega) = \sum_{e\in E} c_e\|\omega_e\|_2.$$

If edges are exchange-equivalent, the weights are constant on each edge orbit. If, in addition, a singleton-edge normalization is imposed,

$$j_e(x) = \|x\|_2 \quad \text{on every normalized edge type,}$$

then $c_e = 1$ on those edge types and

$$J_{2,1}(\omega) = \sum_{e \in E} \|\omega_e\|_2.$$

Proof. By Theorem 3.1, for each edge e there exists $c_e > 0$ such that

$$j_e(x) = c_e \|x\|_2.$$

Hypothesis (2) declares that the global diagnostic is a sum of local edge terms:

$$J(\omega) = \sum_{e \in E} j_e(\omega_e).$$

No independent derivation of separability is claimed in this theorem. Substitution gives the weighted mixed norm:

$$J(\omega) = \sum_{e \in E} c_e \|\omega_e\|_2.$$

If the observer declares edge-exchange equivalence, then isomorphic edge data must receive the same singleton diagnostic, so c_e is constant on each declared edge orbit. This gives an orbit-weighted mixed norm. If the observer further declares singleton-edge normalization $j_e(x) = \|x\|_2$, then each normalized edge has $c_e = 1$. Thus the unweighted $\ell^1(E; \ell^2)$ form is not automatic; it is exactly the normalized single-orbit endpoint. □

Remark (what is, and is not, claimed in Theorem 3.2). Theorem 3.1 pins each local factor to $c_e \| \cdot \|_2$. The outer summation $J(\omega) = \sum_e j_e(\omega_e)$ is *not* a consequence of rotational symmetry and is not newly proved here; it is the explicit strong aggregation declaration in hypothesis (2). Support lemma **S3** is a separate finite theorem: under its stronger zero and disjoint-support splitting hypotheses it can derive an edgewise sum. Weaker aggregate-statistic readings remain outside both results. Theorem 3.2 is therefore a conditional substitution statement, not an independent aggregation theorem.

Thus, $\mathrm{SO}(k)$ selects the local ℓ^2 factor, while the declared strong edgewise rule supplies the outer ℓ^1 sum. The unweighted expression is licensed only under edge normalization or equivalent single-orbit scaling data.

4. Primal Quotient Obstruction and Analytical Duality

In this section and §5, transports are the identity and repairs use $(d_0\alpha)_e = \alpha(v) - \alpha(u)$. Local adjustment fields $\alpha \in C^0(G, \mathbb{R}^k)$ determine the **Quotient Obstruction** $\Phi_{2,1,c}$. For nonidentity transports the analogous repair operator is $(d_{0,r}\alpha)_e = \alpha(v) - r_{v,u}\alpha(u)$, and its adjoint replaces the ordinary divergence in the dual; the simple untransported cycle-sum formula is not asserted in that setting.

[DEFINITION] Definition 4.1 (Primal Form): For positive edge weights $c_e > 0$, define

$$J_{2,1,c}(\eta) := \sum_e c_e \|\eta_e\|_2$$

and

$$\Phi_{2,1,c}([\omega]) := \inf_{\alpha \in C^0(G, \mathbb{R}^k)} \sum_e c_e \|\omega_e - (d_0\alpha)_e\|_2.$$

The normalized case is $c_e = 1$ for all e, written $\Phi_{2,1}$.

The quotient here is $C^1 / \operatorname{im} d_0$. When faces are present, it is not the cohomology space $H^1 = \ker d_1 / \operatorname{im} d_0$: nonclosed cochains are allowed, and Γ records their closure defect. On the cycle graph without 2-cells, $d_1 = 0$ and this quotient agrees with graph H^1.

[STANDARD CONVEX DUALITY / PROVEN FINITE APPLICATION] Theorem 4.2 — Mixed Vector Obstruction Duality Let G be a finite directed graph with identity transports, edge defects in $\mathbb{R}^k$, and positive edge weights $c_e > 0$. The primal quotient obstruction satisfies:

$$\Phi_{2,1,c}([\omega]) = \max_{\varphi} \sum_e \langle \varphi_e, \omega_e \rangle$$

Subject to explicit structural bounds:

1. Gauge Invariance / Divergence-Free Constraint: $d_0^* \varphi = 0$
2. Weighted Euclidean Witness Bounds: $\|\varphi_e\|_2 \le c_e$ for all edges $e \in E$.

Proof. Write the primal problem as

$$\inf_{\alpha} \sum_e c_e \|\omega_e - (d_0 \alpha)_e\|_2.$$

The convex conjugate of the function

$$x \mapsto c_e \|x\|_2$$

is the indicator function of the closed Euclidean ball

$$\{\varphi_e : \|\varphi_e\|_2 \le c_e\}.$$

Using

$$c_e \|r_e\|_2 = \max_{\|\varphi_e\|_2 \le c_e} \langle \varphi_e, r_e \rangle,$$

the primal becomes

$$\inf_{\alpha} \max_{\|\varphi_e\|_2 \le c_e} \sum_e \langle \varphi_e, \omega_e - (d_0 \alpha)_e \rangle .$$

Because all spaces are finite-dimensional, the objective is proper, convex, and continuous, and the primal is feasible. Fenchel–Rockafellar duality permits exchanging infimum and supremum. The α-dependent term is

$$- \langle d_0^* \varphi, \alpha \rangle .$$

The infimum over unconstrained α is finite exactly when

$$d_0^* \varphi = 0.$$

Otherwise the infimum is $-\infty$. Therefore admissible dual witnesses are precisely divergence-free vector circulations with weighted Euclidean bounds. Thus

$$\Phi_{2,1,c}([\omega]) = \max_{\substack{d_0^* \varphi = 0 \\ \|\varphi_e\|_2 \le c_e}} \sum_e \langle \varphi_e, \omega_e \rangle.$$

□

Both extrema are attained. The repair image is a closed finite-dimensional subspace. Positive weights make $J_{2,1,c}$ a norm, so its sublevel sets on the affine residual class are compact and contain a minimizer; a repair field exists for that residual, although it need not be unique. The dual feasible set is a closed bounded intersection of edge balls with $\ker d_0^*$, contains zero, and is compact. Its linear objective therefore attains a maximum. These attainment arguments supplement the continuity qualification used for strong duality above.

In the normalized case $c_e = 1$, the dual constraints reduce to

$$\|\varphi_e\|_2 \le 1.$$

Thus unweighted Euclidean-unit-bounded circulations are exactly the edge-normalized endpoint,

not the general weighted case.

5. Finite Triangle-Anomaly Separation

To isolate the observer-level distinction, we identify a concrete finite cycle defect.

Consider a minimal closed triangle network $V = \{0, 1, 2\}$ with directed edges $E = \{(0, 1), (1, 2), (2, 0)\}$. Let the defect signal over the edges be:

- $\omega_{01} = (1, 0)$
- $\omega_{12} = (0, 1)$
- $\omega_{20} = (0, 0)$

The total cycle circulation evaluates to:

$$c = \omega_{01} + \omega_{12} + \omega_{20} = (1, 1)$$

[STANDARD — graph cycle-space / H^1 of a single cycle; finite exactness kernel Lean-mechanized] For a single directed cycle with identity restrictions, the quotient class is determined by the total circulation

$$S(\omega) = \sum_{e \in C} \omega_e.$$

Indeed, exact repairs $d_0\alpha$ have zero cycle sum, and any two cochains with the same cycle sum differ by an exact repair (the cochains of fixed circulation form a single coset of $\operatorname{im} d_0$ on a one-cycle graph).

Explicitly, label the cycle edges $e_i = (v_i \to v_{i+1})$, with vertex indices modulo n. For $q = \omega - \eta$ with $\sum_i q_{e_i} = 0$, set $\alpha(v_0) = 0$ and $\alpha(v_i) = \sum_{r=0}^{i-1} q_{e_r}$ for $1 \le i < n$. Then $(d_0\alpha)_{e_i} = q_{e_i}$ on every nonclosing edge, and the closing equation follows from the zero sum. If stored edge directions are reversed, use the signed traversal circulation $\sum_i \varepsilon_i \omega_{e_i}$ with $\varepsilon_i \in \{\pm 1\}$; reorienting the cochain reduces to the same construction.

The finite algebraic core of this bridge is mechanized in Lean as `same_circulation_iff_difference_anchored_cycle` and `same_circulation_iff_anchored_cycle_coset`: after choosing an anchored cycle presentation with one closing edge, two cochains have the same circulation exactly when their pointwise difference is an anchored zero-circulation cycle repair, equivalently when they lie in the same anchored repair coset. The remaining prose-standard step is only the identification of that anchored presentation with the usual graph coboundary notation for a concrete cycle.

[PROVEN, CONDITIONAL + single cycle + normalized equal edge weights + norm gauge] Lemma 5.1 (Cycle quotient norm). Let C be a finite directed cycle with edge-local norm j and normalized edge weights. For any vector-valued cochain ω,

$$\Phi_j([\omega]) = j(S(\omega)).$$

Proof. Let $c = S(\omega)$. Every representative η of the same quotient class satisfies $S(\eta) = c$. By the triangle inequality,

$$\sum_{e \in C} j(\eta_e) \ge j\left(\sum_{e \in C} \eta_e\right) = j(c).$$

Conversely, place all circulation on a single edge: choose $\eta_{e_0} = c$ and $\eta_e = 0$ for $e \ne e_0$. This representative has cycle sum c and cost $j(c)$. Therefore the quotient infimum equals $j(c)$. □

The normalized normed-additive kernel of this argument is mechanized in Lean as `normalized_cycle_quotient`.

[PROVEN, CONDITIONAL + single cycle + positive weights] Weighted refinement. If the edge weights are $w_e > 0$, the same argument gives

$$\Phi_{j,w}([\omega]) = \left(\min_{e \in C} w_e\right) j(S(\omega)).$$

The lower bound follows from $\sum_e w_e j(\eta_e) \geq w_{\min} \sum_e j(\eta_e) \geq w_{\min} j(S(\eta))$, and the upper bound is achieved by placing all circulation on an edge of minimal weight. Thus clean numerical comparison requires equal normalized edge weights. The weighted finite-cycle refinement is mechanized in Lean as `weighted_cycle_quotient`.

[PROVEN, NUMERICAL / DIRECT COMPUTATION] Theorem 5.2 (Finite triangle-anomaly separation). On the normalized triangle above,

$$\Phi_{1,1}([\omega]) = 2, \qquad \Phi_{2,1}([\omega]) = \sqrt{2}.$$

Hence coordinate-separable scalar observers and SO(2)-invariant vector observers assign different obstruction magnitudes to the same finite cycle class.

> **Algebraic Extension Note on $\sqrt{2}$.**
>
> The two magnitudes are on distinct scalar footings. $\Phi_{1,1} = 2$ is a rational scalar obtained by summing integer-valued cochain components. $\Phi_{2,1} = \sqrt{2}$ is irrational and requires at minimum the quadratic field extension $\mathbb{Q}(\sqrt{2})$ over the rationals.
>
> In the formalization in `lean4/gs002_l003_01_triangle_separation.lean`, this boundary is verified rationally by proving the squared inequality $2^2 \neq 2$ (that is, $4 \neq 2$) in $\mathbb{Q}$, certifying that the coordinate ℓ^1 cost and SO(2) Euclidean cost are separated without requiring the non-constructive continuum square root.
>
> Minimal replay:

```python
import math

omega01 = (1, 0)
omega12 = (0, 1)
omega20 = (0, 0)

c = tuple(a + b + d for a, b, d in zip(omega01, omega12, omega20))

phi_11 = abs(c[0]) + abs(c[1])
phi_21 = math.sqrt(c[0]**2 + c[1]**2)

print(c)
print(phi_11)
print(phi_21)
```

Expected output:

```
(1, 1)
2
1.4142135623730951
```

This replay checks only the normalized triangle quotient values. It does not certify the general duality theorem or the replica-extensive aggregation theorem.

Proof. The cycle sum is

$$c = (1,0) + (0,1) + (0,0) = (1,1).$$

For the coordinate-separable scalar observer, the local gauge is

$$j_{1,1}(x,y) = |x| + |y|.$$

By Lemma 5.1,

$$\Phi_{1,1}([\omega]) = j_{1,1}(c) = |1| + |1| = 2.$$

For the SO(2)-invariant vector observer, the local gauge is

$$j_2(x,y) = \|(x,y)\|_2.$$

Again by Lemma 5.1,

$$\Phi_{2,1}([\omega]) = j_2(c) = \sqrt{1^2 + 1^2} = \sqrt{2}.$$

Since $2 \neq \sqrt{2}$, the finite triangle separates the observer classes. □

Prediction A — Scalar-Coordinate ℓ^1 ($J_{1,1}$)

If the topological observer separates variables (e.g. tracking independent coordinate channels): $J_{1,1}(\omega) = \sum_e (|\omega_{e,x}| + |\omega_{e,y}|)$ yields a quotient obstruction of:

$$\Phi_{1,1} = \|c\|_1 = |1| + |1| = 2$$

Prediction B — Vector Mixed Norm ($J_{2,1}$)

If the topological observer operates under continuous SO(k) rotational invariance: $J_{2,1}(\omega) = \sum_e \|\omega_e\|_2$ yields a quotient obstruction of:

$$\Phi_{2,1} = \|c\|_2 = \sqrt{1^2 + 1^2} = \sqrt{2} \approx 1.414$$

Finite Numerical Consequence

[**PROVEN, CONDITIONAL on normalization**] The two observers assign different quotient norms to the same cycle class.

An observer declaring separable coordinates ($J_{1,1}$) — the *component-counting* reading — assigns the class magnitude **2**. An observer declaring SO(k)-invariant vector stalks ($J_{2,1}$) — the *Euclidean-distance* reading — assigns the same class magnitude $\sqrt{2}$.

Both numbers are computed from declared gauges on declared data. Naming the first a "component-counting anomaly detector" and the second a "structural strain sensor" describes *candidate interpretations* of the two observer classes; neither is an instrument reading, neither measures a physical system, and no claim is made that any physical detector realizes either gauge (register row **I8**).

This distinguishes observer classes at the level of local gauge choice; it does not by itself test replica-extensive aggregation or dual structure.

The weighted refinement is important. If the scalar-coordinate observer uses edge weights a_e and the SO(2) observer uses edge weights b_e, then the triangle values become

$$\Phi_{1,1,a} = 2 \min_e a_e, \qquad \Phi_{2,1,b} = \sqrt{2} \min_e b_e.$$

Thus the canonical 2 versus $\sqrt{2}$ separation is a normalized comparison. Without normalization, the theorem shows observer-class dependence together with explicit scale dependence, not an absolute numerical invariant.

6. Closure Invariant Bound Extension

Under topological evaluation, the 2-cell failure metric inherits the declared mixed geometry. Let

$$C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} C^2, \qquad d_1 d_0 = 0$$

be a finite obstruction complex with $C^1 = \bigoplus_{e\in E} \mathbb{R}^k$ and $C^2 = \bigoplus_{f\in F} \mathbb{R}^{m_f}$. Equip C^1 with

$$\|\eta\|_{1,2,c} := \sum_{e\in E} c_e \|\eta_e\|_2$$

and equip C^2 with a declared mixed closure norm

$$\|\zeta\|_{1,2,u} := \sum_{f\in F} u_f \|\zeta_f\|_2, \qquad c_e, u_f > 0.$$

Define

$$\Phi_{2,1,c}([h]) := \inf_{\alpha} \|h - d_0\alpha\|_{1,2,c}$$

and

$$\Gamma_{2,1,u}([h]) := \|d_1 h\|_{1,2,u}.$$

The closure diagnostic is well-defined on quotient classes because $d_1 d_0 = 0$.

[PROVEN, CONDITIONAL + finite complex + $d_1 d_0 = 0$ + declared mixed norms] Theorem 6.1 — Mixed Closure Bound. With the definitions above,

$$\Gamma_{2,1,u}([h]) \le \|d_1\|_{(1,2,c)\to(1,2,u)}\, \Phi_{2,1,c}([h]).$$

Proof. Since $d_1 d_0 = 0$, for every repair field α,

$$d_1(h - d_0\alpha) = d_1 h.$$

By the operator norm definition,

$$\|d_1 h\|_{1,2,u} = \|d_1(h - d_0\alpha)\|_{1,2,u} \le \|d_1\|_{(1,2,c)\to(1,2,u)}\, \|h - d_0\alpha\|_{1,2,c}.$$

Taking the infimum over all α gives

$$\Gamma_{2,1,u}([h]) \le \|d_1\|_{(1,2,c)\to(1,2,u)} \Phi_{2,1,c}([h]).$$

□

In the normalized case $c_e = u_f = 1$, this reduces to the unweighted bound

$$\Gamma_2([h]) \le \|d_1\|_{(2,1\to 2,1)} \Phi_{2,1}([h]).$$

This shows the mixed closure bound — the $\Phi \to \Gamma$ link of the $\Phi\Gamma R_{\mathrm{cl}}$ flow — holds for Euclidean rotational fields, with the operator constant carrying the declared edge and face normalization data.

The closure ratio $R_{cl} = \Gamma/\Phi$ is defined only when $\Phi > 0$. When $\Phi = 0$, the ratio is not used; the bound is the $\Phi \to \Gamma$ inequality itself.

For the implementation's common stalk dimension and scalar signed incidence matrix $A = d_1$, the operator constant has the explicit formula

$$\|d_1\|_{(1,2,c)\to(1,2,u)} = \max_{e\in E} \frac{\sum_{f\in F} u_f |A_{fe}|}{c_e}.$$

The triangle inequality bounds each face norm by $\sum_e |A_{fe}| \|h_e\|_2$, giving the upper bound after weighted summation. A cochain supported on a maximizing edge attains it. This formula concerns scalar incidence acting on equal-dimensional stalks; the abstract theorem also allows the more general declared linear operator and face dimensions specified above.

7. Conclusion

Continuous $\mathrm{SO}(k)$ invariance makes the local gauge Euclidean up to scale. The separately declared edgewise sum supplies the outer ℓ^1 form, with weights fixed by singleton normalization or other declared scaling data. Edge exchange symmetry alone supplies common weights on each orbit. In the flat finite graph setting, the resulting quotient norm has the stated dual witnesses, single-cycle formula, and closure bound. The contribution is this explicit separation of hypotheses and their finite obstruction consequences.

7.1 Connection to General Obstruction Theory

The $\ell^1(E;\ell^2)$ geometry derived here integrates into general finite obstruction theory as a specific gauge selection within the broader family of M-compatible gauges:

Observer passivity. In general obstruction models, observers are contractive quotient maps satisfying $\Phi_{\mathcal{O}}(\mathcal{O}(q)) \le \Phi(q)$. When Φ is computed using the weighted or normalized $\ell^1(E;\ell^2)$ geometry, the observer passivity bound constrains how $\mathrm{SO}(k)$-measured defects can be processed by downstream observer maps: no observer quotient can create obstruction absent from the rotationally gauged cochain core.

Contextual distinction. The testable triangle prediction (§5) should be read as an observer-class separation: separable-coordinate and $\mathrm{SO}(k)$-invariant observers assign different quotient magnitudes to the same cycle class (2 versus $\sqrt{2}$). It illustrates that quotient defect magnitudes depend essentially on declared local measurement symmetry rather than reflecting a single observer-independent metric.

Sheaf and network gluing. The target-stalk defect $\delta_e(a) = a(v) - r_{v,u}a(u)$ represents local mismatch. Declared $\mathrm{SO}(k)$-invariant local gauges and the strong edgewise sum give the weighted or normalized $\ell^1(E;\ell^2)$ diagnostic. Theorem 4.2 supplies ordinary divergence-free witnesses for identity transports. With nonidentity transports, dual witnesses instead satisfy the adjoint constraint for $d_{0,r}$; no untwisted circulation formula or holonomy lower bound is supplied here.

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